WRRA Full-State Junction: Development Audit v0.7-dev-r1

Won sik Choi · Jeongin Choi

2026-10-20

1. Verified inputs

The frozen inputs are the 64 evolved v0.4 modes with 13 neutrino moments, the v0.5 finite-packet clock with 256 reservoir partitions, and the v0.6 massive-electron thermal completion. The original 128-cell seed is used without amplitude refitting. The inherited WRRA dark-weight-only target is $\omega_c = 0.120378061773 78$. This target and known physical constants are explicitly declared calibration inputs.

2. WRRA transformations

The complete available scalar state of each v0.4 mode is transformed into the post-exchange CDM-comoving synchronous frame. Density, velocity, metric derivatives and the neutrino monopole/dipole are transformed together. The r1 adapter also stores the small metric derivative combination directly to prevent cancellation loss. Separately, the seed-dependent event-time motion of v0.5 is included in the v0.6 thermal equations through a common-time saltation tangent. These two validated calculations still describe different source models. They are not represented as a single completed finite-k integration.

3. Computed outputs

Quantity	Result
Maximum transformed Hamiltonian constraint residual	3.29×10^{-16}
Maximum momentum constraint residual	4.31×10^{-16}
Maximum absolute entropy gauge-invariance error	$< 1.12 \times 10^{-16}$
Finite-packet thermal $k = 0$ entropy response	0.194818034056
Earlier smooth-source response	0.723567385056
Original-seed entropy RMS under the new local tangent	4.80263×10^{-9}
Thermally normalized packet ω_c at old β	0.103663511129
Relative discrepancy against inherited late target	-13.885047%
Recalibrated conversion fraction β	$8.85389093544 \times 10^{-7}$
Change in conversion fraction	+16.123841%

The old packet normalization produces about 13.9% less present dark matter than the inherited downstream input. This is a junction normalization discrepancy, not an observational exclusion of the WRRA structure. Recalibration yields $\omega_c = 0.120378062005$. This is legitimate background calibration, not an independent density prediction. The 64 finite-k modes have not yet been rerun on that new background.

4. Falsification and checks

Both Einstein constraints, inverse transformation, entropy invariance and retention of high neutrino moments are tested mode by mode. Simply deleting the CDM velocity violates the momentum constraint by about 1.47%. Omitting event-time motion incorrectly gives zero response for this model.

Independent central differences of nonlinear histories agree with the saltation entropy tangent to at most 2.68×10^{-7} relatively. Simultaneously refined EOS grid, quadrature and time tolerance change the response by 1.39×10^{-12} relatively. Energy-plus-pressure-work residuals remain below 3.11×10^{-11} and 2.53×10^{-12} for the base and refined runs. All 472 acceptance checks pass.

The derivative is valid only on the fixed-count open branch $-0.00281226035 < \xi < 0.00845879466$. All original 128 cells lie inside it. These checks establish the stated numerical transformations, not an observational validation of a completed cosmology.

5. Conclusion and remaining junction

Step 4 remains partially implemented. Executed advances are the conserving 64-mode frame transformation, local thermal/event-time tangent and background recalibration. No new CMB spectrum or lensing likelihood is calculated here.

A genuine next integration must specify the finite-k energy/momentum transfer across each event hypersurface; complete baryon/electron and massive-neutrino distribution states; rerun modes on the recalibrated thermal background; and declare the three-dimensional/angular covariance of the finite seed. The first two requirements cannot be supplied by rearranging arrays. An angular completion can be a declared conditional model, but none is introduced here. Exporting these partial states as complete CAMB initial conditions raises an explicit error listing the unmet physical requirements.

Equations and conventions

For $y = (N, \ln T_\gamma, M, W)$ and $N = \ln a_{\rm rel}$, C is constant between events. The background equations are

$$\rho = \rho_{\rm EM}(T) + \rho_{\nu,i} e^{-4N} + C e^{-4N} + M e^{-3N},$$

$$\dot{N} = H, \qquad \dot{\ln T} = -\frac{3H(\rho_{\rm EM} + p_{\rm EM})}{T \partial_T \rho_{\rm EM}}, \qquad \dot{M} = 0.$$

Dots in the implemented ODE use the inherited dimensionless time. The finite electron mass, thermal Fermi integrals, effective $N_{\rm eff} = 3.044$ input, and instantaneous relativistic neutrino decoupling branch are inherited from v0.6. This is not a collision-integral calculation or a BBN yield prediction.

At event j, with $q = \beta \rho_i / 256$ and $\epsilon = 0.01$,

$$C^+ = C^- - q, \quad M^+ = M^- + (1-\epsilon) q e^{-N}, \quad \rho_{\rm EM}(T^+) = \rho_{\rm EM}(T^-) + \epsilon q e^{-4N}.$$

The rate modulation is the original seed ξ, with no amplitude refit. In the fixed-count neighborhood $t_j(\xi) = t_j(0)/(1+\xi)$. For the common-time tangent $z = \partial y / \partial \xi$, the reset is

$$z^+ = DR\, z^- + (DR\, f^- - f^+) t_j', \qquad t_j' = -t_j/(1+\xi).$$

Between events $\dot{z} = (\partial f / \partial y) z$. The last term above is essential. Merely differentiating the state reset loses the entire response in this autonomous fixed-packet model. On the final common proper-time slice,

$$S_{m,\rm EM} = \frac{z_M}{M} - 3 z_N - \frac{\partial_{\ln T} \rho_{\rm EM}}{\rho_{\rm EM} + p_{\rm EM}} z_{\ln T}.$$

This expression is a $k = 0$ tangent of local cells. It supplies no new finite-gradient radiation or momentum source.

For the separate frozen v0.4 smooth-source states, a residual synchronous transformation into the post-exchange CDM frame is

$$\alpha = \theta_m / k^2, \quad \theta_i^c = \theta_i - k^2 \alpha, \quad \delta_i^c = \delta_i + 3(1+w_i) H \alpha, \quad \eta^c = \eta + H\alpha.$$

Here $\alpha' = -H\alpha$ after exchange. The implementation also transforms h' and η', with $\eta^{c\prime} = \eta' + (H' - H^2)\alpha$ and $h^{c\prime} = h' + 2k^2\alpha - 6(H' - H^2)\alpha$. The neutrino monopole and dipole change; all retained moments $\ell \geq 2$ remain intact. CAMB conventions are etak $= k\eta$, $q_\gamma = 4\theta_\gamma/(3k)$, and $q_\nu = F_{\nu 1}$. The adapter exports a partial native variable dictionary, not a complete CAMB evolution array. No metric h is invented from an unspecified spatial gauge integration constant.

Release revalidation and numerical correction

Ten supplementary release checks pass in addition to the 472 base checks. The independent clock coordinate $u = (1+\xi)t$ keeps events fixed and integrates $w' = Jw - f$, $w^+ = DRw^-$; the final derivative is $z = w + t_{\rm end} f$. It recovers the entropy response within relative 8.39×10^{-9}. A five-point numerical flow Jacobian agrees within 1.78×10^{-10}.

Reconstructing $h' + 6\eta'$ from its separately rounded transformed terms causes up to 16.66% error in the small Bardeen potential at the lowest mode. The r1 adapter therefore retains the combination directly as $h' + 6\eta' = 2k^2(g/2 + \alpha)$, using the inherited scaled shear g. The Bardeen potential computed from this stable combination is gauge-invariant within relative 4.35×10^{-12} over all 64 modes. The lossy reconstruction is kept as a

negative control. Separate derivative components are diagnostics; they must not be used to recover this small difference. A future actual CAMB transport must also verify stable constraint evaluation; this release does not claim to have performed that transport.

The original entropy response, target calibration and 472 base checks are unchanged. An independent SI-unit reconstruction yields $\omega_c = 0.103663511046343$, consistent with the thermal-density calculation.

Sources and reproducibility

- WRRA v0.4: DOI 10.5281/zenodo.23490141.
- WRRA v0.5: DOI 10.5281/zenodo.23491041.
- WRRA v0.6: DOI 10.5281/zenodo.23491673.
- Ma and Bertschinger (1995), astro-ph/9506072: scalar gauge and Einstein-Boltzmann conventions.
- CAMB 2.0.4 pristine local source, `fortran/equations.f90`: native variable conventions; scalar initial-state assignment and constraint equations. This artifact does not modify or run CAMB.
- CODATA constants through SciPy 1.17.0; the same finite-mass electromagnetic EOS code as the frozen v0.6 input.

Run `python run_all.py` with the pinned requirements. The bundle includes all inputs used here, scripts, machine-readable outputs, checks and hashes. The original v0.4/v0.5/v0.6 publications remain unchanged. Release DOI: 10.5281/zenodo.23492328. No new CMB or lensing result is claimed.